

WEEK 4 - ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

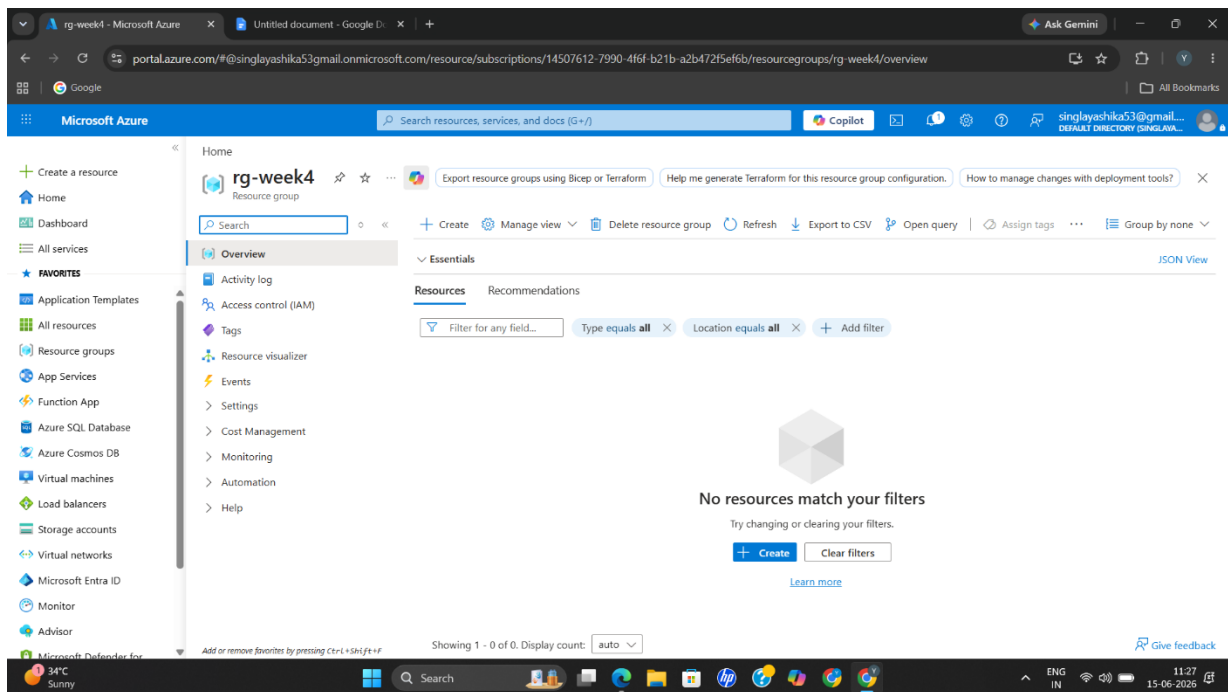
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



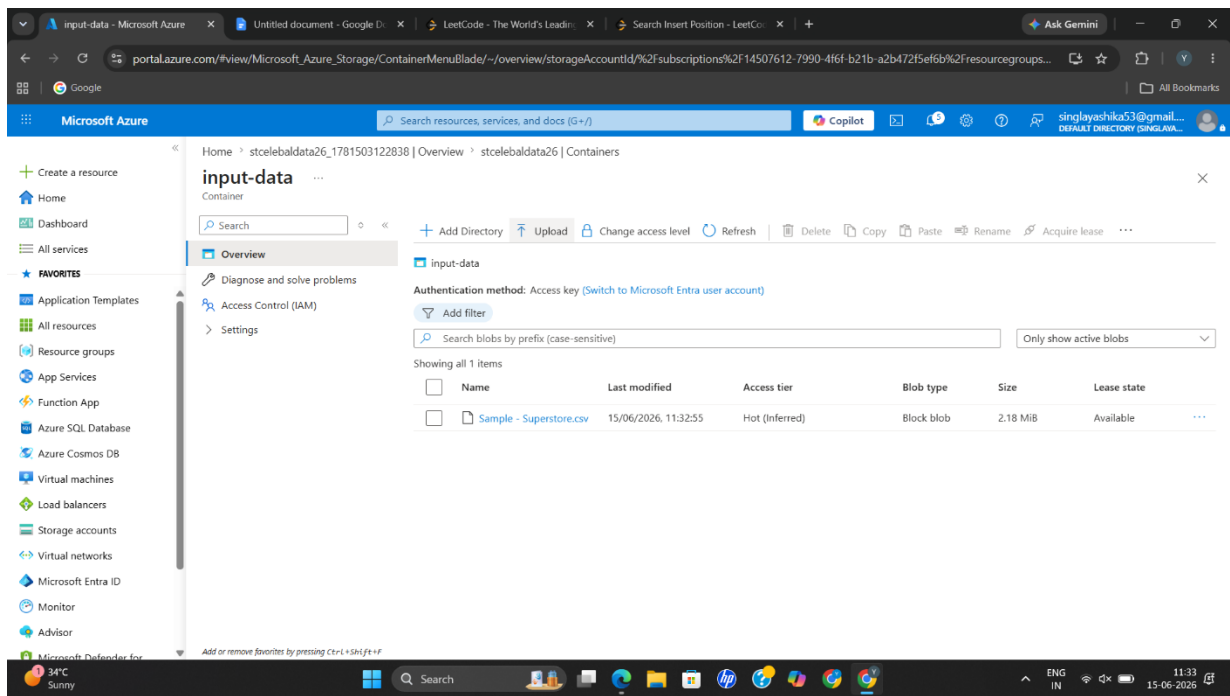
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file



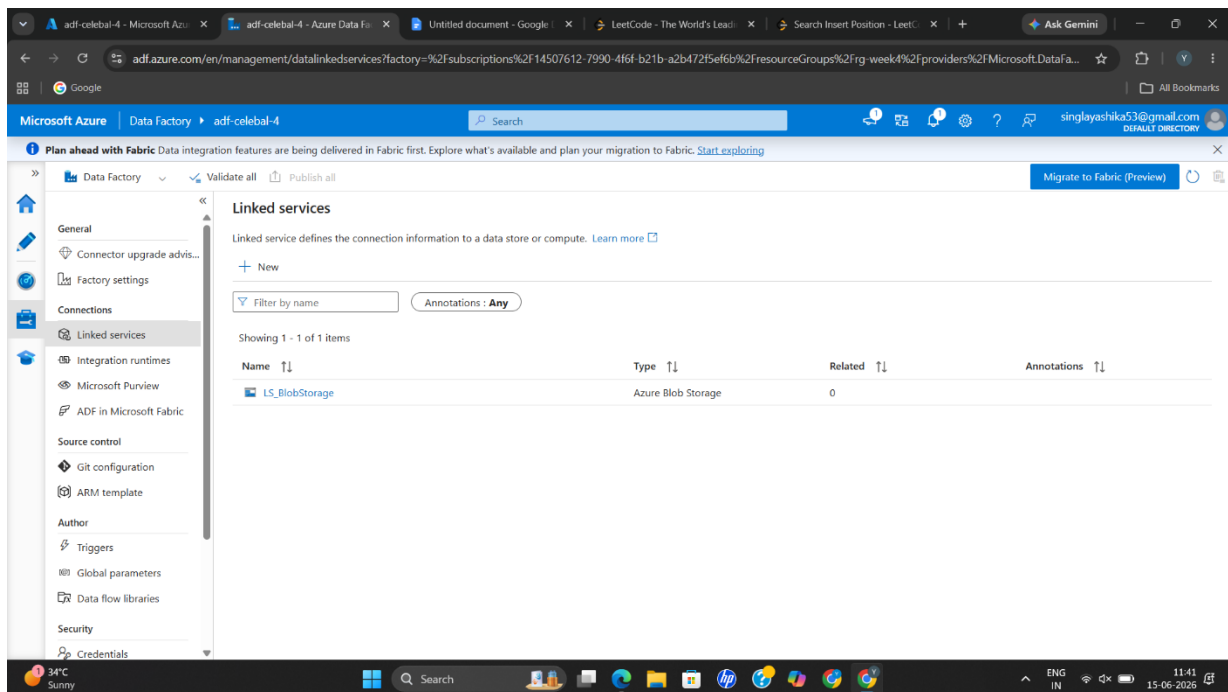
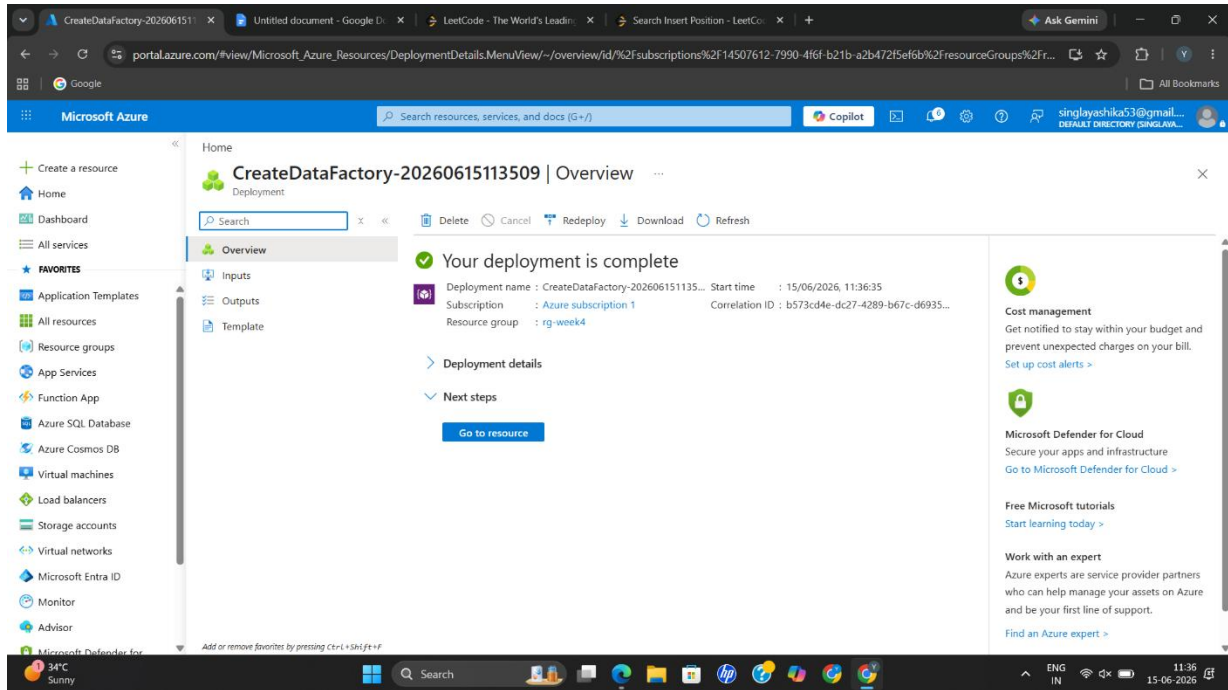
Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service



- Screenshot of dataset

The image displays two screenshots of the Microsoft Azure Data Factory portal, showing the configuration of datasets within a factory named 'adf-celebal-4'.

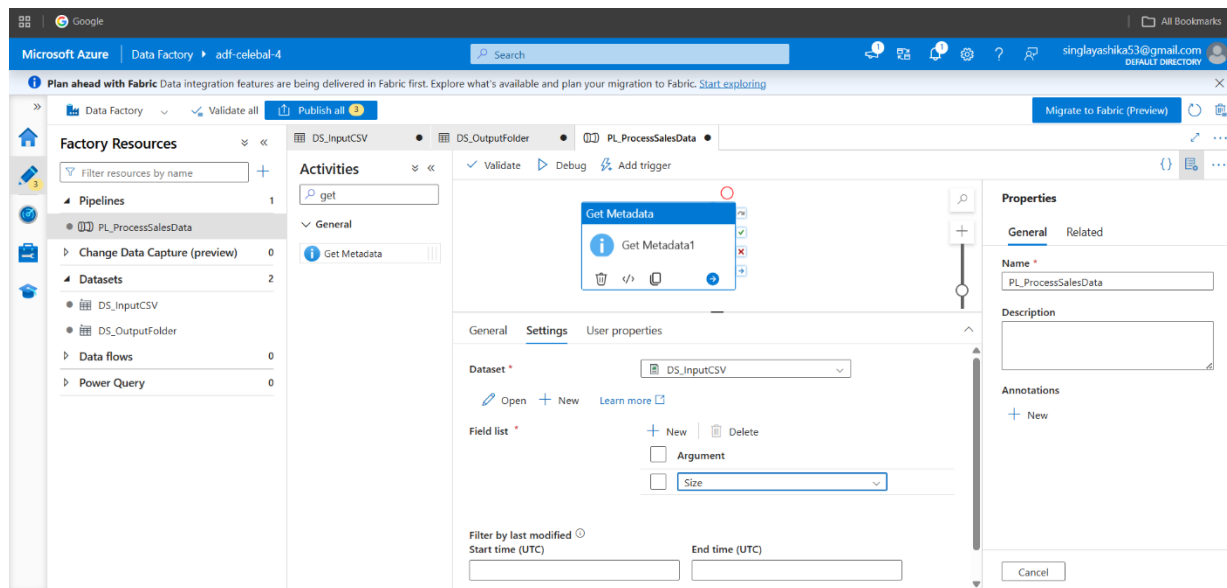
Top Screenshot: DS_InputCSV Dataset Configuration

- Factory Resources:** The left sidebar shows the hierarchy: Data Factory > Datasets > DS_InputCSV.
- Dataset Details:** The main pane shows the 'DS_InputCSV' dataset, which is a 'DelimitedText' type.
- Connection Tab:** The 'Connection' tab is active, showing the following settings:
 - Linked service:** LS_BlobStorage
 - File path:** input-data / Directory / Sample - Superstore.csv
 - Compression type:** No compression
 - Column delimiter:** Comma (,)
 - Row delimiter:** Default (\n, or \r\n)
 - Encoding:** Default(UTF-8)
 - Quote character:** Double quote (")
 - Escape character:** Backslash (\)
- Properties Tab:** The 'Properties' tab is visible on the right, showing the dataset name 'DS_InputCSV' and a description field.

Bottom Screenshot: DS_OutputFolder Dataset Configuration

- Factory Resources:** The left sidebar shows the hierarchy: Data Factory > Datasets > DS_OutputFolder.
- Dataset Details:** The main pane shows the 'DS_OutputFolder' dataset, which is a 'DelimitedText' type.
- Connection Tab:** The 'Connection' tab is active, showing the following settings:
 - Linked service:** LS_BlobStorage
 - File path:** output-data / Directory / File name
 - Compression type:** No compression
 - Column delimiter:** Comma (,)
 - Row delimiter:** Default (\n, or \r\n)
 - Encoding:** Default(UTF-8)
 - Quote character:** Double quote (")
 - Escape character:** Backslash (\)
- Properties Tab:** The 'Properties' tab is visible on the right, showing the dataset name 'DS_OutputFolder' and a description field.

- Screenshot of Get Metadata activity



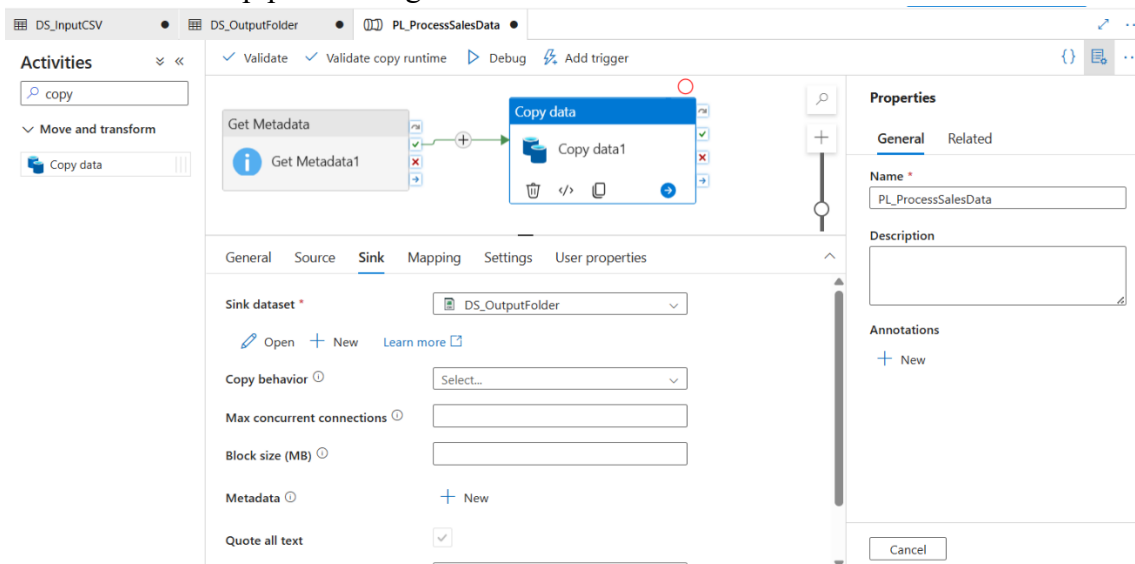
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



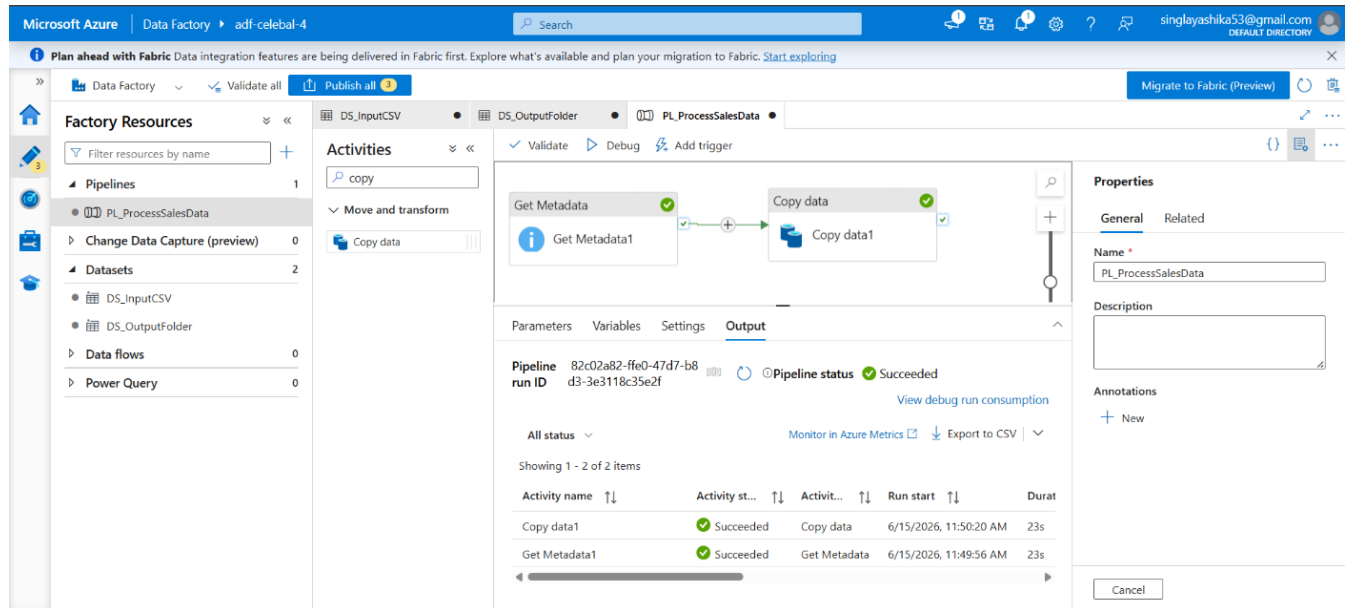
Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



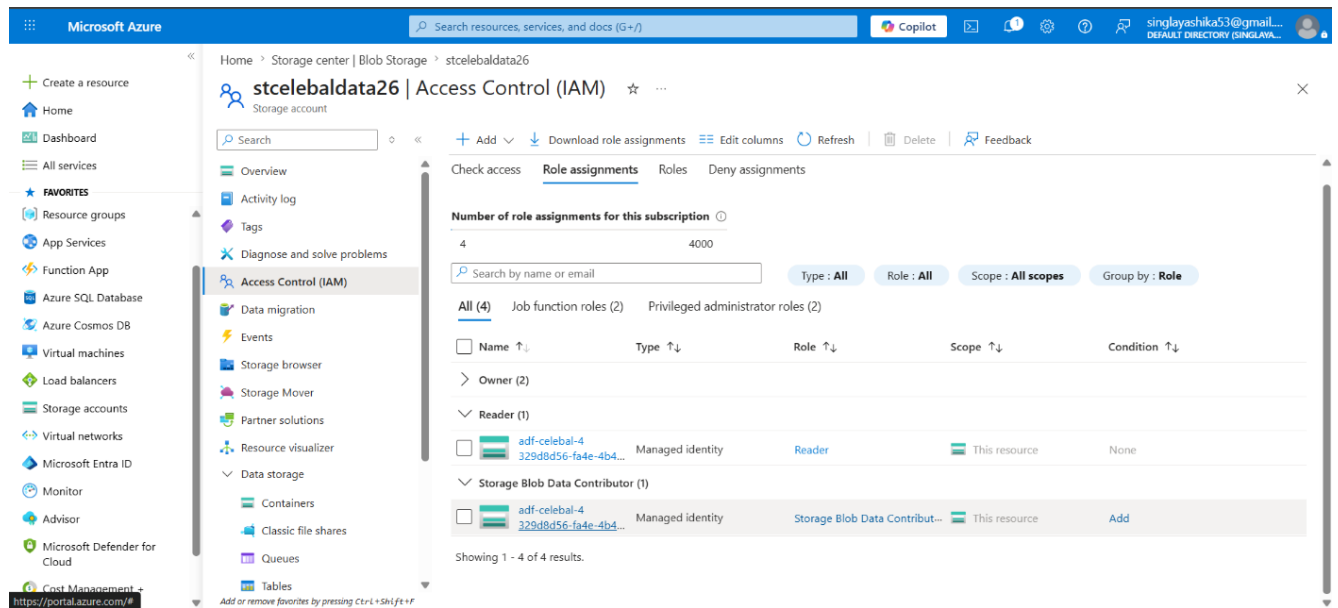
Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully

Microsoft Azure | Data Factory | adf-celebal-4

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines 1
 - PL_ProcessSalesData
- Datasets 2
 - DS_InputCSV
 - DS_OutputFolder
- Data flows 0
- Power Query 0

Activities

- copy
- Move and transform
 - Copy data

Get Metadata1 → Copy data1

Parameters Variables Settings **Output**

Pipeline run ID: 82c02a82-ffe0-47d7-b8d3-3e3118c35e2f Pipeline status: Succeeded

View debug run consumption

All status: Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Durat
Copy data1	Succeeded	Copy data	6/15/2026, 11:50:20 AM	23s
Get Metadata1	Succeeded	Get Metadata	6/15/2026, 11:49:56 AM	23s

Properties

General Related

Name: PL_ProcessSalesData

Description:

Annotations: + New

Cancel

- Data copied to destination

DS_InputCSV DS_OutputFolder PL_ProcessSalesData

Activities

- copy
- Move and transform
 - Copy data

Get Metadata1 → Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset: DS_OutputFolder

Copy behavior: Select...

Max concurrent connections:

Block size (MB):

Metadata: + New

Quote all text: ☒

Properties

General Related

Name: PL_ProcessSalesData

Description:

Annotations: + New

Cancel

- Metadata validated

Micr

i

>>

Output

Copy to clipboard

```
{
  "size": 2287806,
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 7
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
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        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```